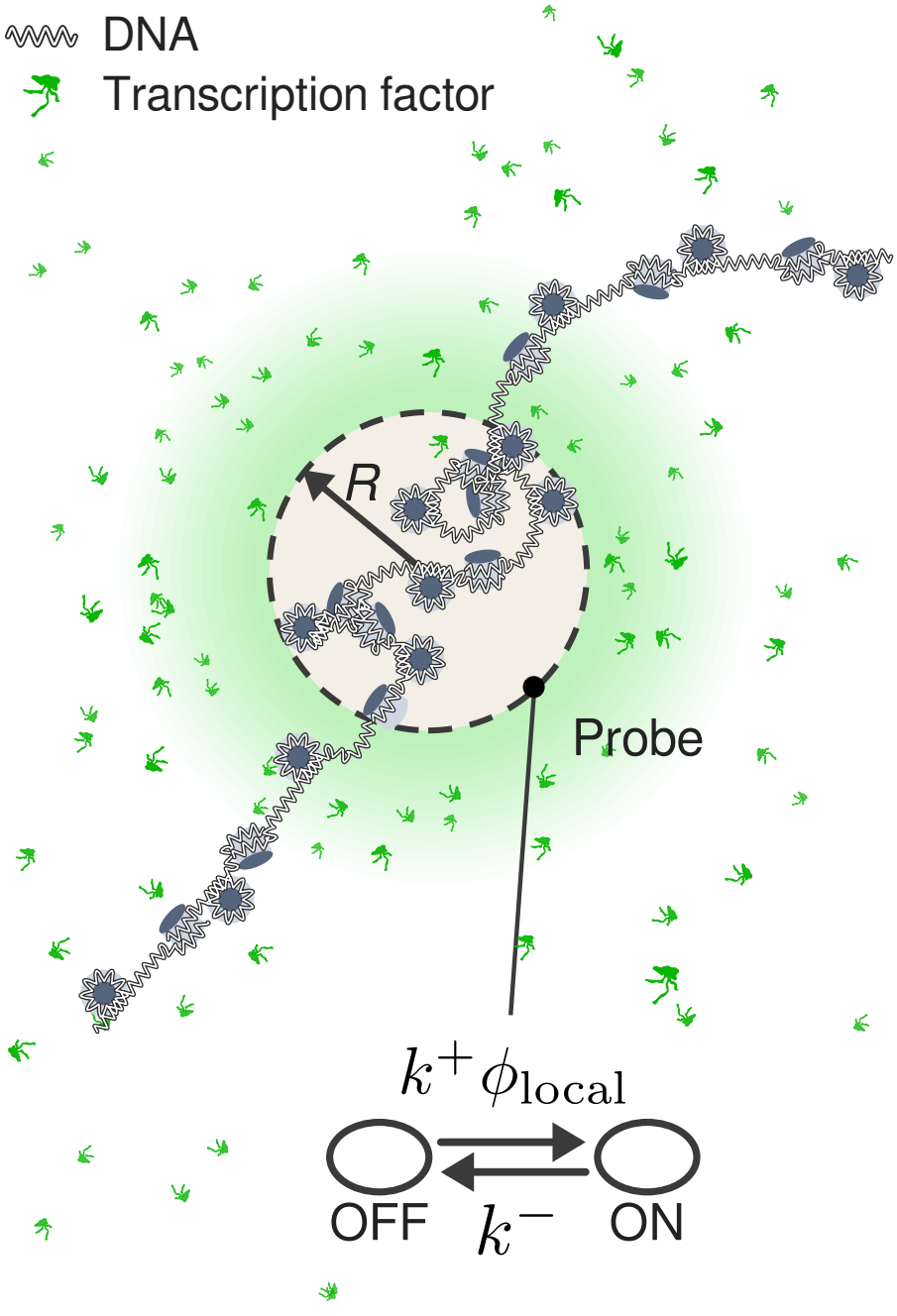


Free receptor

$$\dot{\mathcal{F}}_{\text{free}}(\phi) = \frac{1}{\tau_{\text{free}}(\phi)}$$

Receptor at the attractive surface

$$\dot{\mathcal{F}}(\phi) = \underbrace{\frac{1}{\tau_{\text{free}}(\phi_{\text{local}})}}_{\text{Probe}} \underbrace{\left(\frac{d\log\phi_{\text{local}}}{d\log\phi}\right)^2}_{\text{Surface}}$$